Boosting

Gradient Boosting: CWB Basics 2

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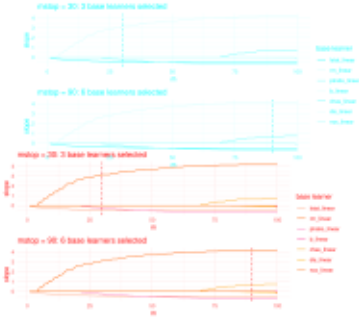

Learning goals

- Handling of categorical features
- Intercept handling
- Practical example

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HANDLING OF CATEGORICAL FEATURES

Feature x_j with G categories. Two options for encoding:

- One base learner to simultaneously estimate all categories:

$$b_j(x_j|\boldsymbol{\theta}_j) = \sum_{g=1}^{G} \theta_{j,g} \mathbb{1}_{\{g=x_j\}} = (\mathbb{1}_{\{x_j=1\}}, \ldots, \mathbb{1}_{\{x_j=G\}})\boldsymbol{\theta}_j$$

  Hence, b_j incorporates a one-hot encoded feature with group means $\boldsymbol{\theta} \in \mathbb{R}^G$ as estimators.

- One binary base learner per category:

$$b_{j,g}(x_j|\theta_{j,g}) = \theta_{j,g} \mathbb{1}_{\{g=x_j\}}$$

  Including all categories of the feature means adding G base learners $b_{j,1}, \ldots, b_{j,G}$

INTERCEPT HANDLING

There are two options to handle the intercept in CWB. In both, the loss-optimal constant $f^{[0]}(\mathbf{x})$ is an initial model intercept.

1. Include an intercept BL:
   - Add BL $b_{\text{int}} = \theta$ as potential candidate considered in each iteration and remove intercept from all linear BLs, i.e., $b_j(\mathbf{x}) = \theta_j x_j$.
   - Final intercept is given as $\hat{f}_0^{[0]}(\mathbf{x}) + \hat{\theta}$. Linear BLs without intercept only make sense if covariates are centered (see Hofner et al. 2014 tutorial, p. 7)
2. Include intercept in each linear BL and aggregate into global intercept post-hoc:
   - Assume linear base learners $b_j(\mathbf{x}) = \theta_{j1} + \theta_{j2} x_j$. If base learner $\hat{b}_j$ with parameter $\hat{\boldsymbol{\theta}}^{[1]} = (\hat{\theta}_{j1}^{[1]}, \hat{\theta}_{j2}^{[1]})$ is selected in first iteration, model intercept is updated to $f^{[0]}(\mathbf{x}) + \hat{\theta}_{j1}^{[1]}$.
   - During training, intercept is adjusted M times to yield $f^{[0]}(\mathbf{x}) + \sum_{m=1}^{M} \hat{\theta}_{j1}^{[m]}$

EXAMPLE: LIFE EXPECTANCY

Consider the `Life expectancy` data set (WHO, available on Kuhn, 2019): regression task to predict life expectancy.

We fit a CWB model with linear BLs (with intercept)

variable	description
`Life.expectancy`	Life expectancy in years
`Country`	The country (just a selection GER, USE, SWE, ZAF, and ETH)
`Year`	The recorded year
`BMI`	Average BMI $= \frac{\text{body weight in kg}}{(\text{Height in m})^2}$ in a year and country
`Adult.Mortality`	Adult mortality rates per 1000 population

Using `compboost` with $M = 150$ iterations, we can visualize which BL was selected when and how the estimated feature effects evolve over time.

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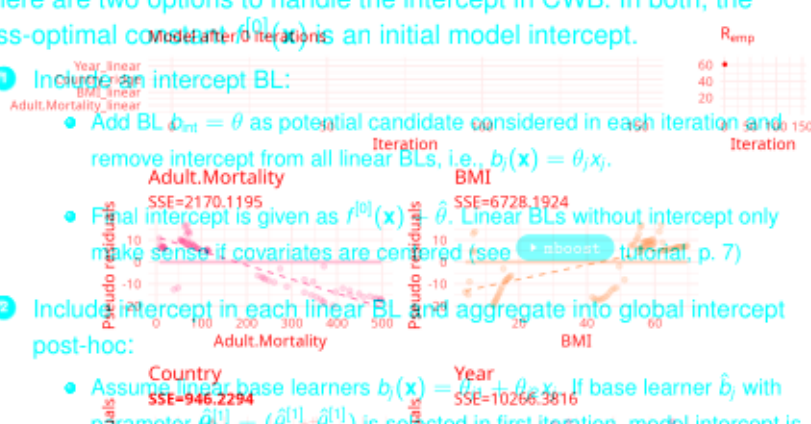

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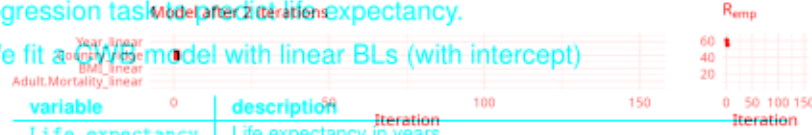

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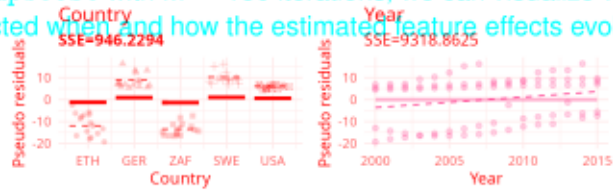

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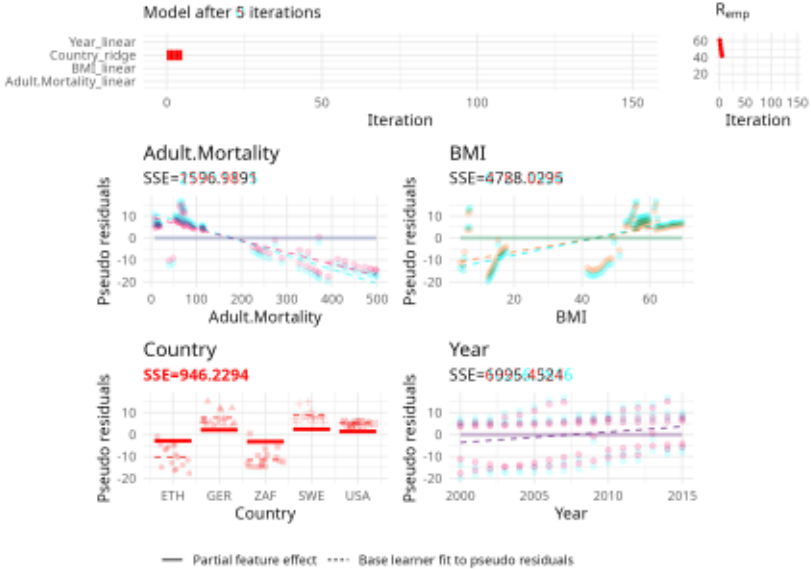

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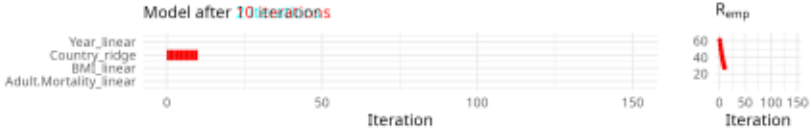

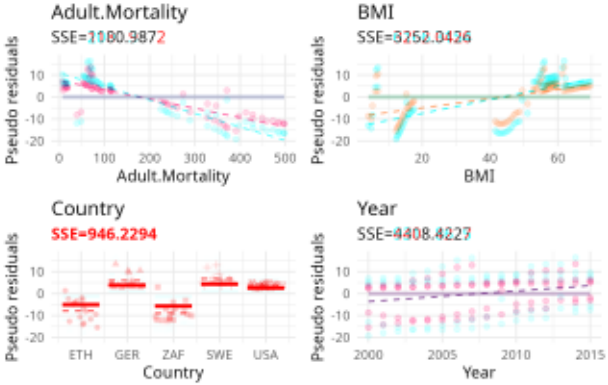

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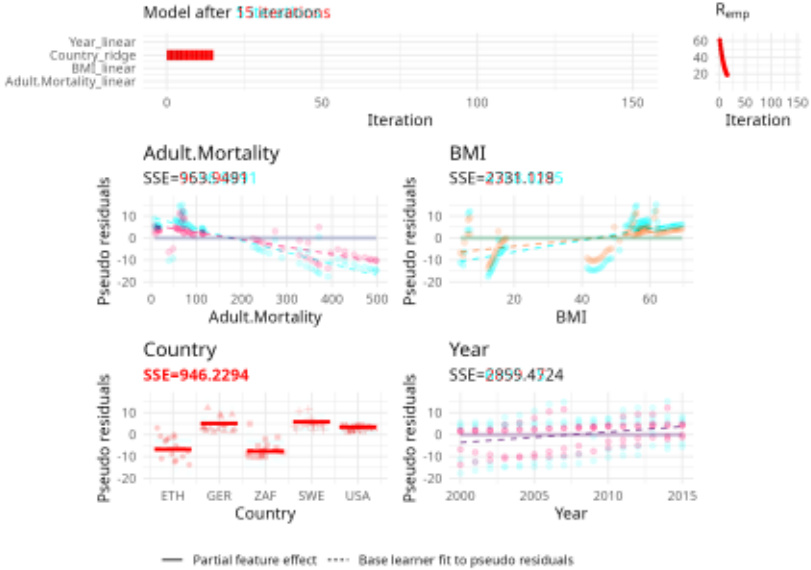

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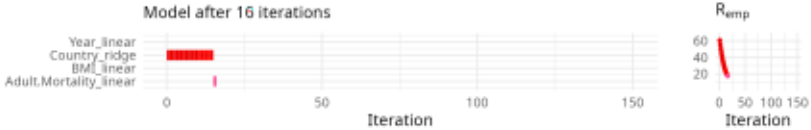

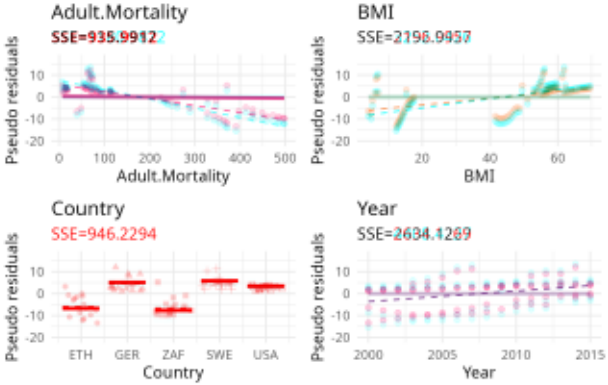

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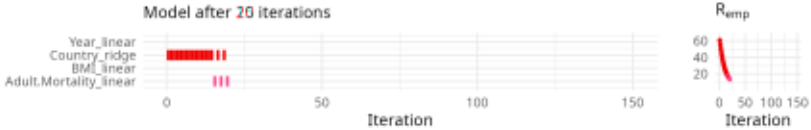

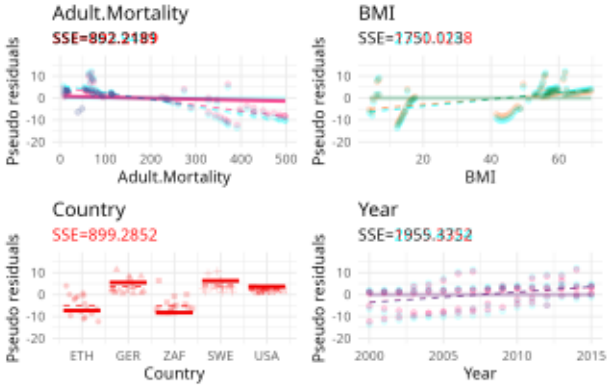

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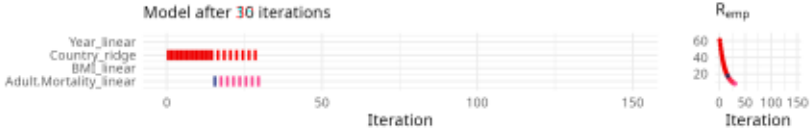

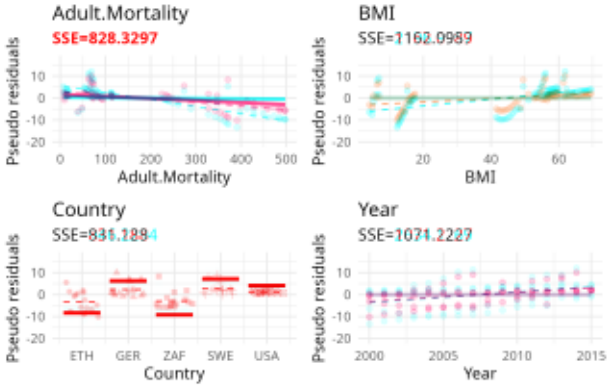

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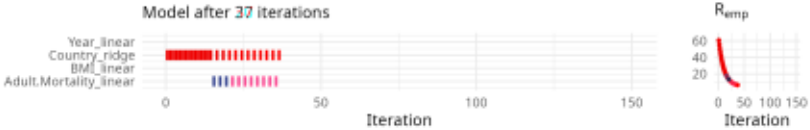

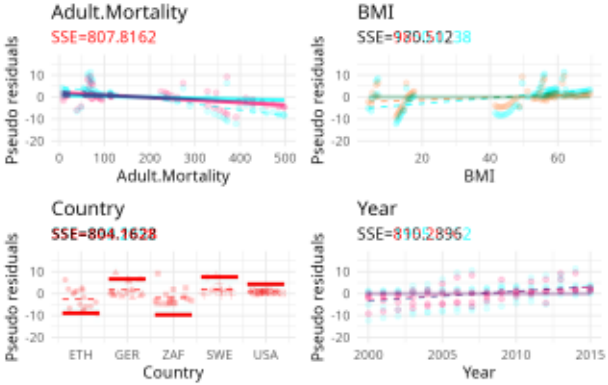

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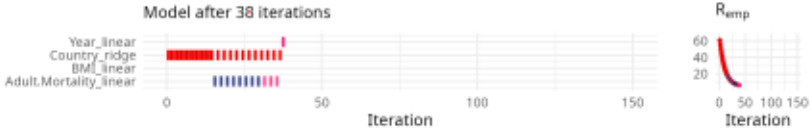

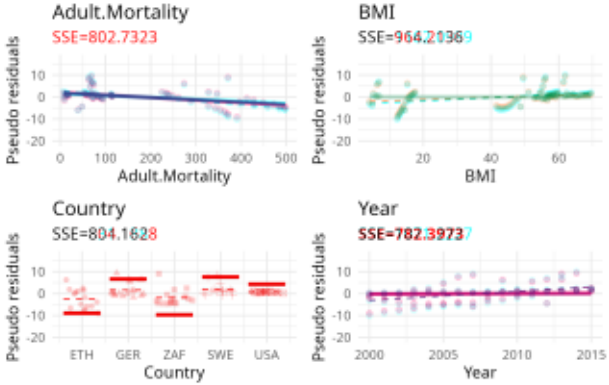

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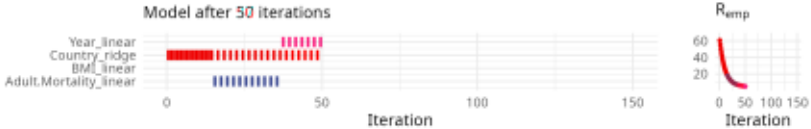

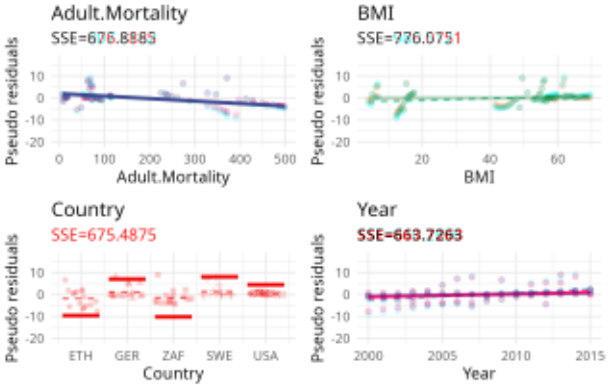

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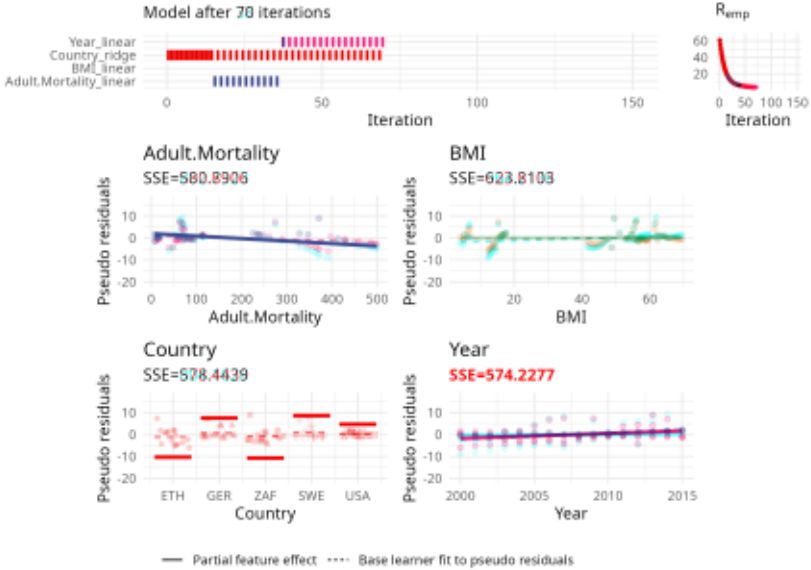

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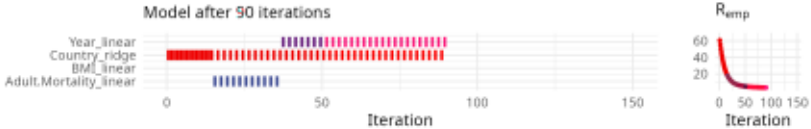

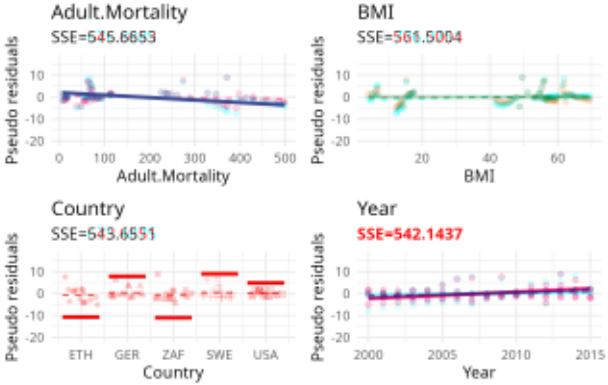

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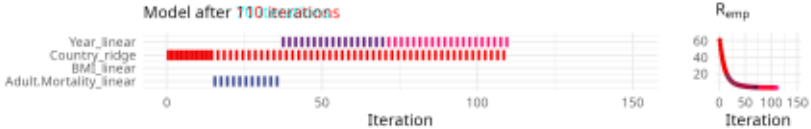

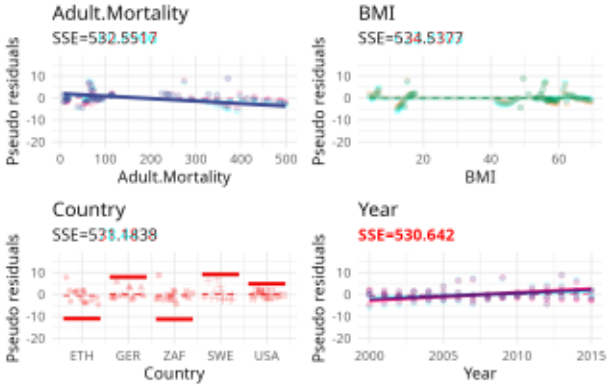

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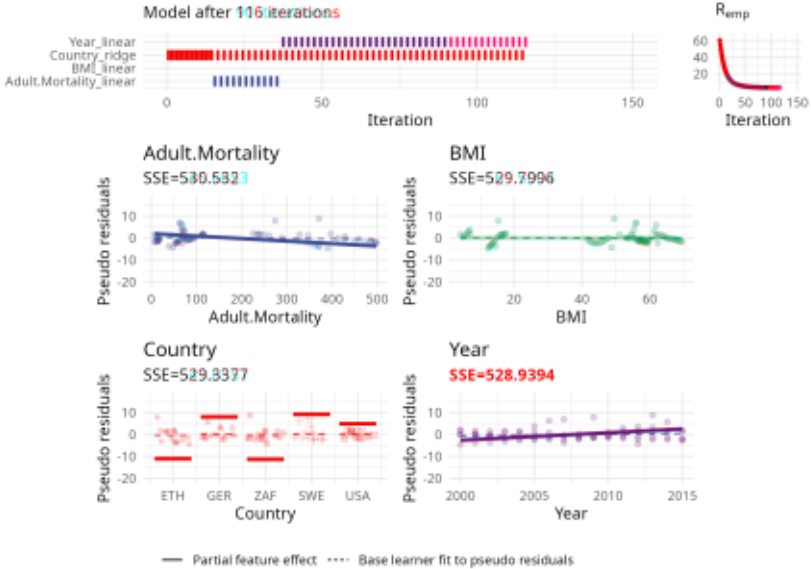

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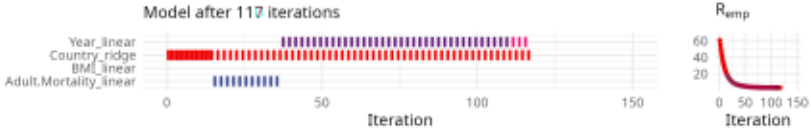

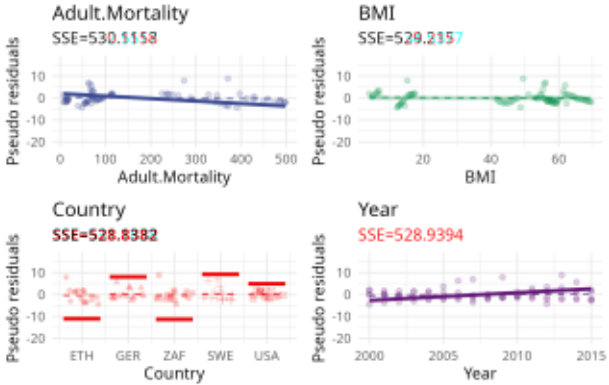

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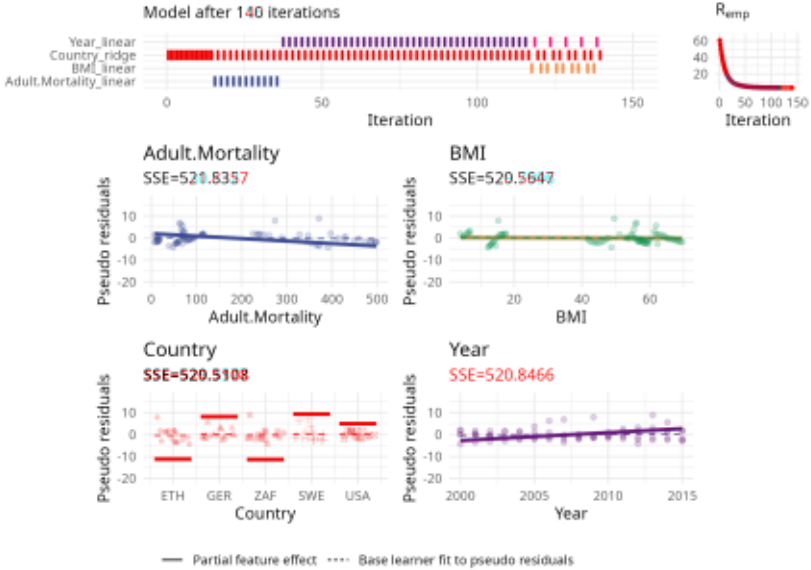

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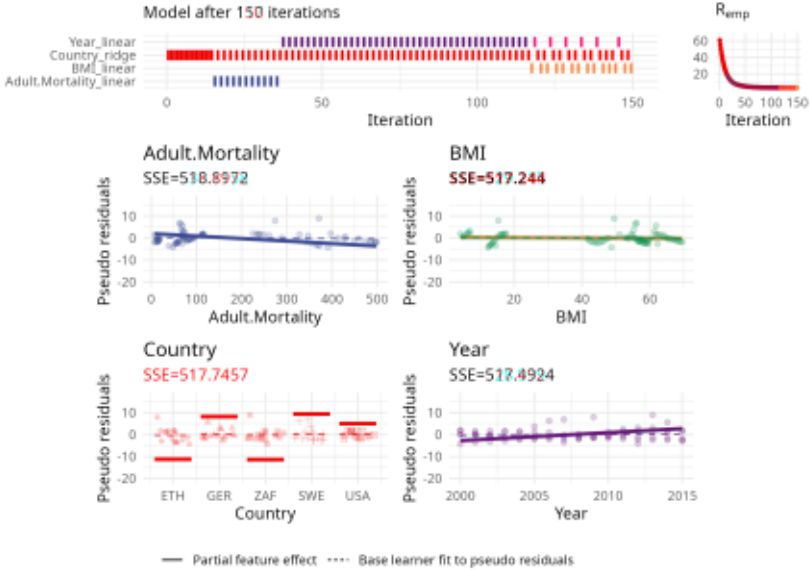

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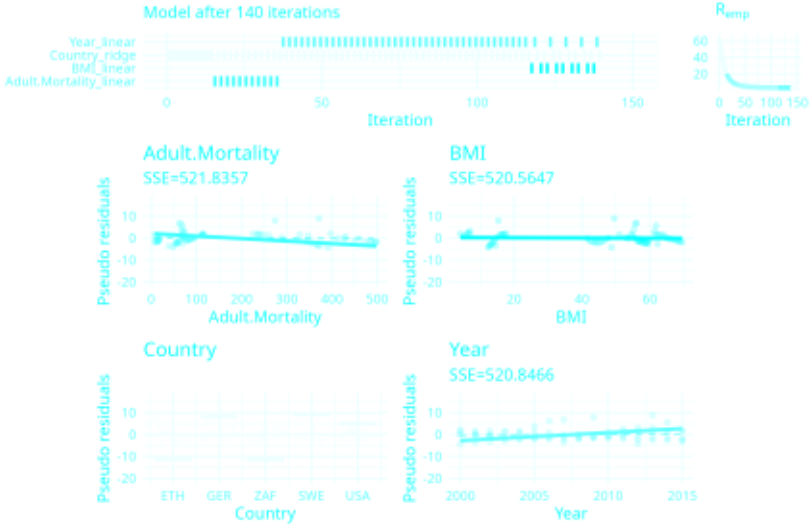
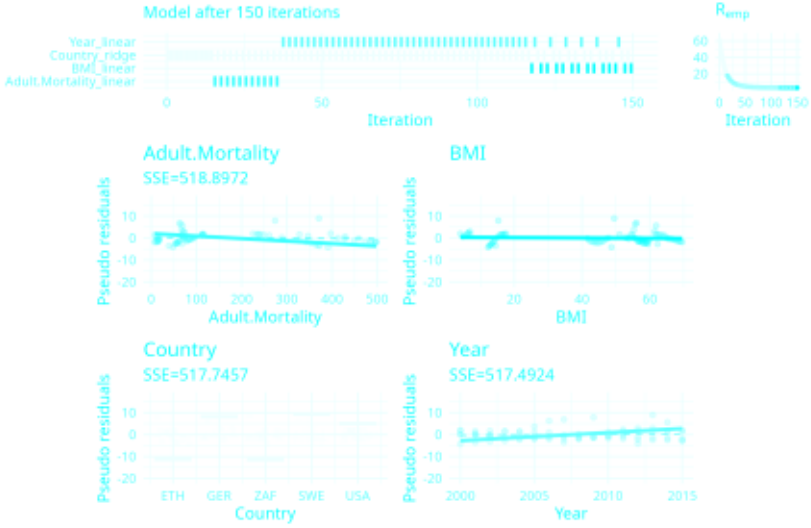
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